

Chapter 1

Introduction

Alzheimer’s disease (AD) diagnosis in practice depends on two kinds of evidence that are usually held apart: brain-imaging scans (MRI or PET) that show structural change, and clinical or demographic records that capture cognitive test scores, genetic risk markers, and lifestyle history. A moderator of AD risk who reads only the scan misses the clinical context that determines how alarming a given degree of atrophy is, and a system that reads only the clinical record misses the structural evidence a scan provides. Both kinds of evidence are, in addition, collected by different hospitals and research centres, and regulatory restrictions on patient data mean this information cannot simply be pooled onto one server for training. This chapter states the resulting problem, the objectives adopted in response, and the contributions this project claims.

1.1 Problem Statement

Alzheimer’s disease is a progressive neurodegenerative disorder whose early and accurate diagnosis remains a major clinical challenge, since reliable detection typically requires the joint analysis of neuroimaging data (MRI/PET) and clinical or demographic risk factors (age, genetic markers, cognitive scores) collected across multiple hospitals and research centres. Centralising this sensitive medical data to train a single large-scale model is often infeasible because of patient privacy regulations, data-sharing restrictions between institutions, and the heterogeneity of imaging protocols and patient populations across sites. Federated Learning (FL) offers a privacy-preserving alternative by allowing multiple clients (hospitals or sites) to collaboratively train a shared model without exchanging raw data [9]. However, existing federated frameworks for Alzheimer’s diagnosis largely rely on naive or random client participation strategies, ignoring differences in data quality, modality completeness, and class distribution across clients [1, 3, 4]. This can lead to biased global models, slower convergence, and reduced diagnostic reliability, especially when pretrained vision foundation models are fine-tuned in such heterogeneous, non-IID federated settings. There is therefore a need for an adaptive client selection mechanism, integrated with a multimodal vision foundation model, that can identify and prioritise the most informative and reliable clients during federated training so as to improve the accuracy, fairness, and efficiency of Alzheimer’s

disease prediction.

1.2 Problem Background

Deep learning applied to neuroimaging has repeatedly shown that convolutional and transformer-based encoders can recover diagnostically useful structure from MRI and PET scans that is difficult to specify by hand [2]. Vision foundation models pretrained at scale on large image or vision-language corpora, such as the contrastive vision-language approach of Radford et al. [8], have extended this further by supplying representations that transfer to a new task with comparatively little labelled data, and recent medical adaptations such as ADLIP [2] apply this idea directly to Alzheimer’s diagnosis by jointly encoding MRI scans and clinical text. These systems, however, are trained in a centralised setting, which is difficult to reconcile with the privacy constraints that govern hospital data.

Federated Learning [9] was proposed precisely to remove the need for centralisation: clients train locally and only model updates, not raw data, are exchanged with a central server. Mohanraj and Radhakrishnan [1] demonstrate that a federated, multimodal approach can reach competitive accuracy for Alzheimer’s diagnosis while preserving privacy, and Mouheb et al. [3] show that a pretrained 3D medical foundation model (SAM-Med3D) can be fine-tuned under federated constraints, with the choice of aggregation strategy and encoder-freezing configuration measurably affecting performance under non-IID data. Parameter-efficient fine-tuning methods such as LoRA [7] make this style of federated fine-tuning practical by updating only a small low-rank subset of parameters, which reduces the size of the updates each client must transmit.

What remains largely unaddressed in this line of work is *how clients are chosen* to participate in a given round of training. The default in almost all of the systems above is random sampling or full participation. The broader federated learning literature shows that this default is not benign: Li, Chen, and Teng [4] survey a wide range of client selection strategies and show that client heterogeneity is a major factor in convergence speed and fairness, and Tang et al. [12] propose FedCor, which uses the correlation structure between client updates to select a subset expected to reduce the global loss fastest. Neither of these general-purpose strategies is designed for the specific constraints of multimodal medical data, where a client hospital may have an MRI scan without a matching clinical record, may serve a population skewed toward a particular disease stage, or may simply have lower image quality than another site. This project is positioned at the intersection

of these two bodies of work: it takes the federated, multimodal, foundation-model pipeline that recent Alzheimer’s-diagnosis work has established, and combines it with a client selection mechanism that is explicitly aware of the data-quality and modality constraints that arise in a real hospital network.

1.3 Research Objectives

- To design a multimodal federated learning framework that fuses neuroimaging (MRI) data with clinical and demographic risk factors for Alzheimer’s disease prediction.
- To integrate a pretrained vision foundation model into the federated pipeline using parameter-efficient fine-tuning (LoRA) [7] so that the communication cost of each round is kept low enough for resource-constrained hospital clients.
- To develop and evaluate an adaptive client selection strategy that scores clients on local data quality, modality completeness, and class balance, and that selects the subset of clients most likely to improve the global model in each round.
- To compare the proposed client-selection-aware framework against standard baselines, including FedAvg [9] with random client sampling and correlation-based selection [12], on diagnostic accuracy, convergence speed, and communication efficiency.
- To make model decisions and client contributions interpretable to clinicians using Grad-CAM [10] for the imaging branch and SHAP [11] for the clinical branch.

1.4 Motivations

Three observations motivated this project. The first is regulatory and practical: hospital data cannot be centralised for training, so any system intended for real deployment across multiple sites has to be federated by design, not as an afterthought. The second observation concerns the data itself. A federation of hospital clients is rarely uniform: sites differ in scanner hardware and imaging protocol, in the demographic composition of their patient population, and in how completely they have recorded clinical variables alongside imaging. Treating every client as equally informative, as random or full-participation sampling does, discards information that is available almost for free once each client can report a quality or completeness score for its own data. The third observation concerns trust. A clinician who is asked

to act on a federated model’s prediction has no way to inspect the training process directly, and if the contribution of an unreliable client cannot be discounted, the resulting prediction is harder to trust than one from a system where client quality is explicitly accounted for and where the prediction itself can be explained through Grad-CAM and SHAP.

1.5 Significance of the Research

If successful, this project contributes a client selection mechanism that is reusable beyond Alzheimer’s diagnosis, since the underlying problem, that federated clients vary in reliability and completeness, is common to most multi-hospital medical AI applications. The communication-efficiency component, built around LoRA fine-tuning of a foundation model, is significant for low-resource hospital settings that cannot afford to transmit full model updates every round. The explainability component addresses a practical adoption barrier: clinicians are considerably more likely to trust and act on a system whose predictions can be traced to specific imaging regions or clinical variables. Finally, by directly comparing an adaptive selection strategy against random sampling and against a general-purpose strategy such as FedCor [12], the project produces evidence on whether medical-domain-aware selection criteria offer a measurable benefit over domain-agnostic ones, which is a question the existing literature has not yet answered for this setting.

1.6 Research Contribution

- A federated learning framework in which a shared vision foundation model encoder and a clinical feature encoder are fused and fine-tuned locally using LoRA, so that only a small set of low-rank parameters is communicated in each round.
- An adaptive client selection algorithm that scores each client on local data quality, modality completeness, and class balance before every aggregation round, in place of the random or full-participation sampling used in prior federated Alzheimer’s-diagnosis work [1, 3].
- A weighted aggregation scheme in which the contribution of each selected client is scaled by its selection score, extending standard FedAvg [9] rather than replacing it outright.
- An explainability layer that pairs Grad-CAM [10] attributions on the imaging branch with SHAP [11] attributions on the clinical branch, so that a flagged

prediction is accompanied by evidence from both modalities.

- A planned ablation study isolating the effect of client selection, of the weighted aggregation scheme, and of each modality in turn, to establish which components of the framework are responsible for any observed improvement over the FedAvg and FedCor baselines.

1.7 Complex Engineering Analysis

This research project may be regarded as an example of a complex engineering problem for three reasons. First, it requires knowledge from several distinct areas at once, transformer-based vision foundation models, parameter-efficient fine-tuning, federated optimisation, and statistical measures of client heterogeneity. Second, it presents conflicting requirements: diagnostic accuracy, communication efficiency, patient privacy, and fairness across heterogeneous clients compete with one another, and improving one can degrade another. Third, its components, client scoring, weighted aggregation, and foundation-model fine-tuning, are mutually interdependent, since a change to the scoring function changes which clients are aggregated, which in turn changes what the shared encoder learns.

1.7.1 Complex Engineering Problems

The technical issues in this context include reconciling heterogeneous imaging and clinical data collected under different protocols and populations at each hospital site; designing a client-quality scoring function that captures data quality, modality completeness, and class balance simultaneously, for which no standard formulation exists; and balancing the conflicting goals of diagnostic accuracy, communication efficiency, and fairness across clients of very different sizes and reliability.

1.7.2 Complex Engineering Activities

The project involves federated optimisation, parameter-efficient fine-tuning of a pretrained vision foundation model, design of a multimodal fusion architecture combining imaging and clinical encoders, an adaptive client-scoring and weighted-aggregation mechanism, and Explainable AI techniques (Grad-CAM and SHAP) to make the resulting predictions interpretable to clinicians. It also requires pre-processing and partitioning of multiple public and structured medical datasets to simulate a realistic hospital federation.

1.8 Report Organization

The remainder of this report is organised as follows. Chapter 2 reviews the literature on deep learning and vision foundation models for Alzheimer’s diagnosis, federated learning for medical applications, client selection strategies in federated learning, and explainability in medical AI, and closes with an analysis of the gap this project addresses. Chapter 3 presents the research methodology: the proposed framework, the constituent datasets and their preparation, the model and algorithm design, and the planned explainability and ablation components. Chapter 4 will report the implementation and result analysis once experiments are conducted. Chapter 5 states the standards, constraints, and ethical considerations that apply to the work, together with the project timeline. Chapter 6 will conclude the report and identify directions for further work.

Chapter Summary

This chapter introduced the proposed adaptive client-selection federated framework for multimodal Alzheimer’s disease prediction. It combines MRI imaging and clinical and demographic risk factors under a federated learning setting, using a vision foundation model fine-tuned with LoRA and an adaptive client selection mechanism that accounts for data quality, modality completeness, and class balance. Explainable AI techniques (Grad-CAM and SHAP) are used to make the resulting predictions interpretable to clinicians.

References

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